Economics 5040: Dynamic Stochastic Modeling
Problem Set 8 — Income Risk, Precautionary Saving, and Self-Insurance

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The central model for this problem set is

$$\max E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$a_{t+1} = Ra_t + z_t - c_t, \qquad a_{t+1} \geq \underline{a},$$

where z_t is stochastic labor income.

Problem 1. Two-Period Saving Under Income Risk

Consider a household with

$$u(c) = \log c, \qquad y_0 = 10, \qquad \beta = 0.95, \qquad R = 1.02.$$

The household chooses saving $a_1 \geq 0$. Consumption is

$$c_0 = 10 - a_1, \qquad c_1 = Ra_1 + Y_1.$$

Future income is

$$Y_1 = \begin{cases} 2 & \text{with probability } 1/2, \\ 10 & \text{with probability } 1/2. \end{cases}$$

(a) Write the household's expected-utility maximization problem.

(b) Derive the first-order condition for an interior optimum. Explain its economic meaning in terms of the marginal cost and expected marginal benefit of saving one additional unit.

(c) Use Python to solve numerically for optimal saving a_1^*. Report c_0 and consumption in each of the two possible second-period states.

(d) Now replace risky income by certain income

$$Y_1 = E[Y_1] = 6.$$

Solve again for optimal saving.

(e) Compare saving under risky and certain income. Explain the difference using the curvature of marginal utility.

(f) Plot expected lifetime utility against a_1 for both the risky-income and certain-income economies on the same graph. Mark the two optimal choices.

(g) Suppose the household could purchase actuarially fair insurance against the second-period income shock. What would you expect to happen to precautionary saving? Explain without doing a numerical calculation.

Problem 2. The Infinite-Horizon Income Fluctuation Problem

Consider an infinitely lived household with CRRA utility

$$u(c) = \frac{c^{1-\rho} - 1}{1 - \rho},$$

with the usual log-utility case when $\rho = 1$.

Income follows a two-state Markov chain,

$$z_t \in \{1, 2\},$$

with transition matrix

$$P = \begin{pmatrix} 0.90 & 0.10 \\ 0.20 & 0.80 \end{pmatrix}.$$

The household faces

$$a_{t+1} = Ra_t + z_t - c_t, \qquad a_{t+1} \geq 0.$$

Use

$$\beta = 0.96, \qquad R = 1.02, \qquad \rho = 2.$$

(a) Identify the state variables, choice variable, and feasible set.

(b) Write the Bellman equation.

(c) Derive the Euler equation when the borrowing constraint does not bind:

$$u'(c_t) = \beta R E_t[u'(c_{t+1})].$$

Write the corresponding condition when the borrowing constraint binds.

(d) Construct an asset grid

$$a \in [0, 20]$$

and solve the model by value-function iteration.

(e) Plot the optimal next-period asset policy $a'(a, z)$ for both income states.

(f) Plot the corresponding consumption policies.

(g) Explain economically why the saving policy differs between the low- and high-income states. In particular, identify where the borrowing constraint appears to matter.

Problem 3. Self-Insurance in Simulation

Use your solution from Problem 2. Start a household with

$$a_0 = 2, \qquad z_0 = 2.$$

Simulate the economy for $T = 1{,}000$ periods.

(a) Plot income, assets, and consumption for the first 100 periods.

(b) Discard the first 200 periods and report the mean and standard deviation of income, consumption, and assets over the remaining periods.

(c) Compare the volatility of consumption with the volatility of income. Does the household smooth consumption?

(d) Find several episodes in which income switches from the high state to the low state. What happens to assets and consumption following the negative income shock?

(e) Repeat the simulation imposing

$$a_{t+1} = 0$$

in every period, so that the household cannot save. Compare consumption volatility in the two economies.

(f) Explain the term *self-insurance* using your numerical results. Be precise about what the household is insuring itself against and what asset holdings accomplish.

Problem 4. Precautionary Saving and the Amount of Risk

Return to the infinite-horizon model in Problem 2.

Consider two income processes with the same conditional mean but different amounts of income risk. Construct a second income process in which the low and high income realizations are closer together than in the benchmark case while preserving the same mean income. Keep β, R, preferences, and the borrowing constraint unchanged.

(a) Solve the household's problem under both income processes.

(b) Plot the asset policy functions from the two economies together.

(c) Simulate a long time series from each economy and compare average asset holdings.

(d) Which economy generates greater desired asset holdings? Explain the result in terms of precautionary saving.

(e) Explain carefully the distinction among these three ideas:

$$\boxed{\text{consumption smoothing}} \qquad \boxed{\text{self-insurance}} \qquad \boxed{\text{precautionary saving}}.$$

Your answer should explain how the three concepts are related without treating them as synonyms.

What surprised me?

In a short paragraph, identify one result from this problem set that you found surprising or initially counterintuitive. Explain why.